

## *E9 205 Machine Learning for Signal Processing*

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# EM Algorithm

## For Mixture Gaussian Models

11-09-2019

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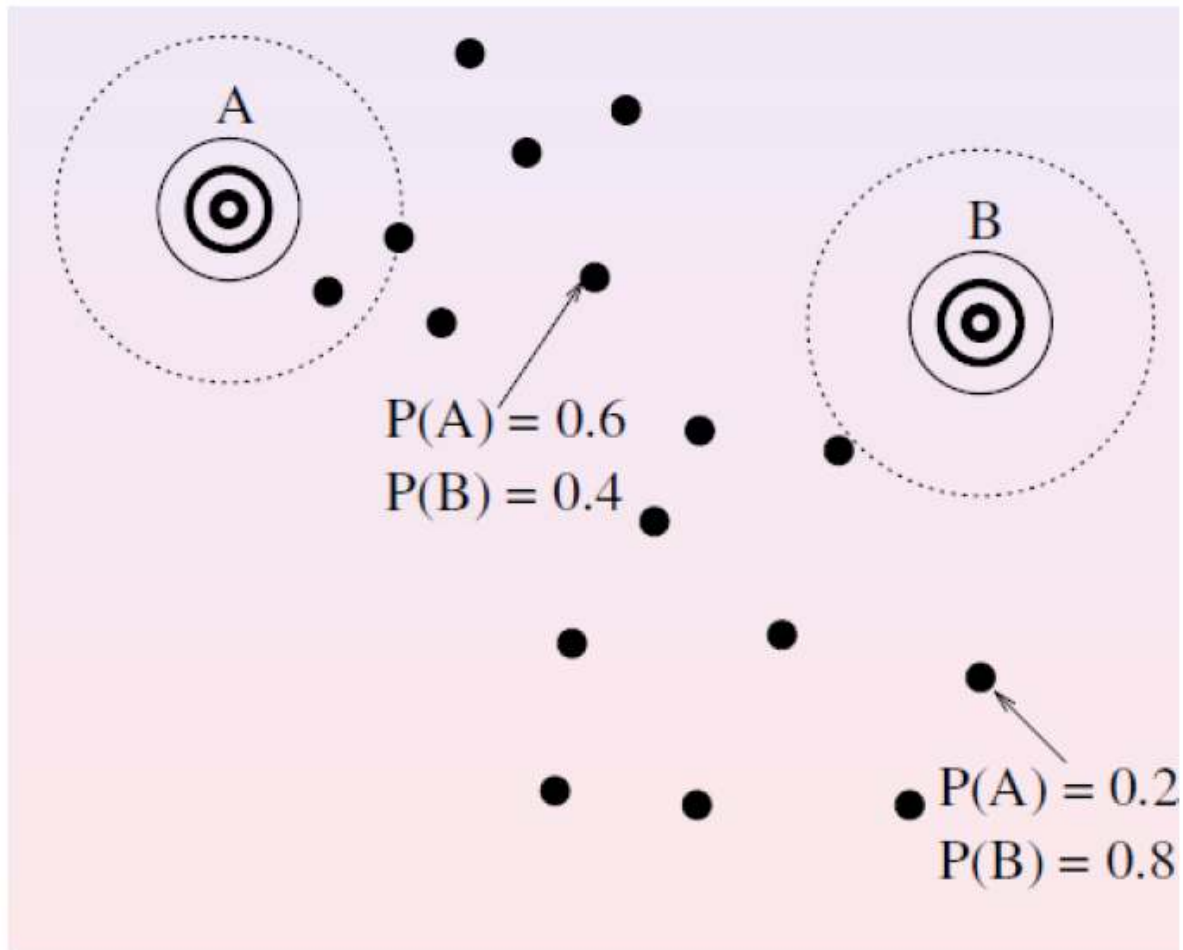
Teaching Assistant - Prachi Singh ([prachisingh@iisc.ac.in](mailto:prachisingh@iisc.ac.in)).



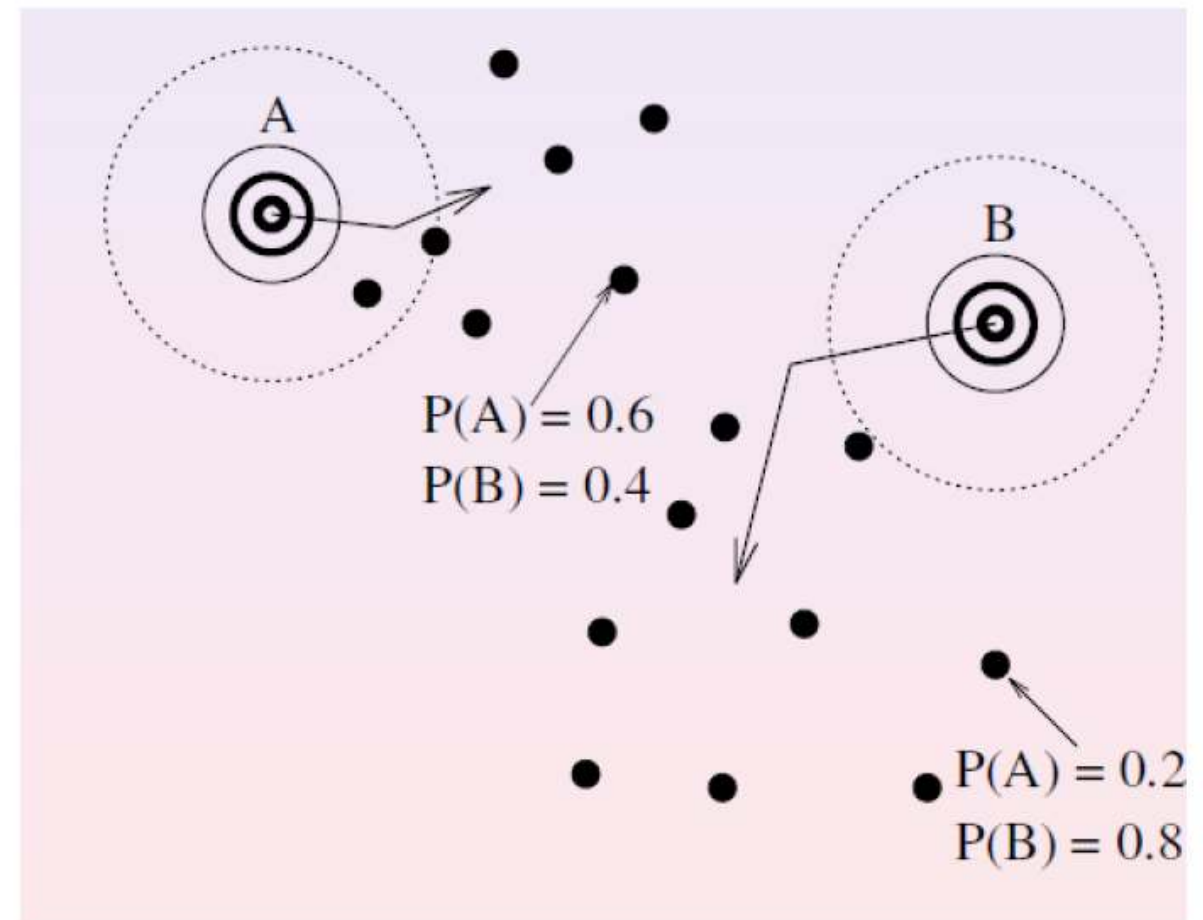
# EM Algorithm For GMMs

# EM Algorithm for GMM

E-step

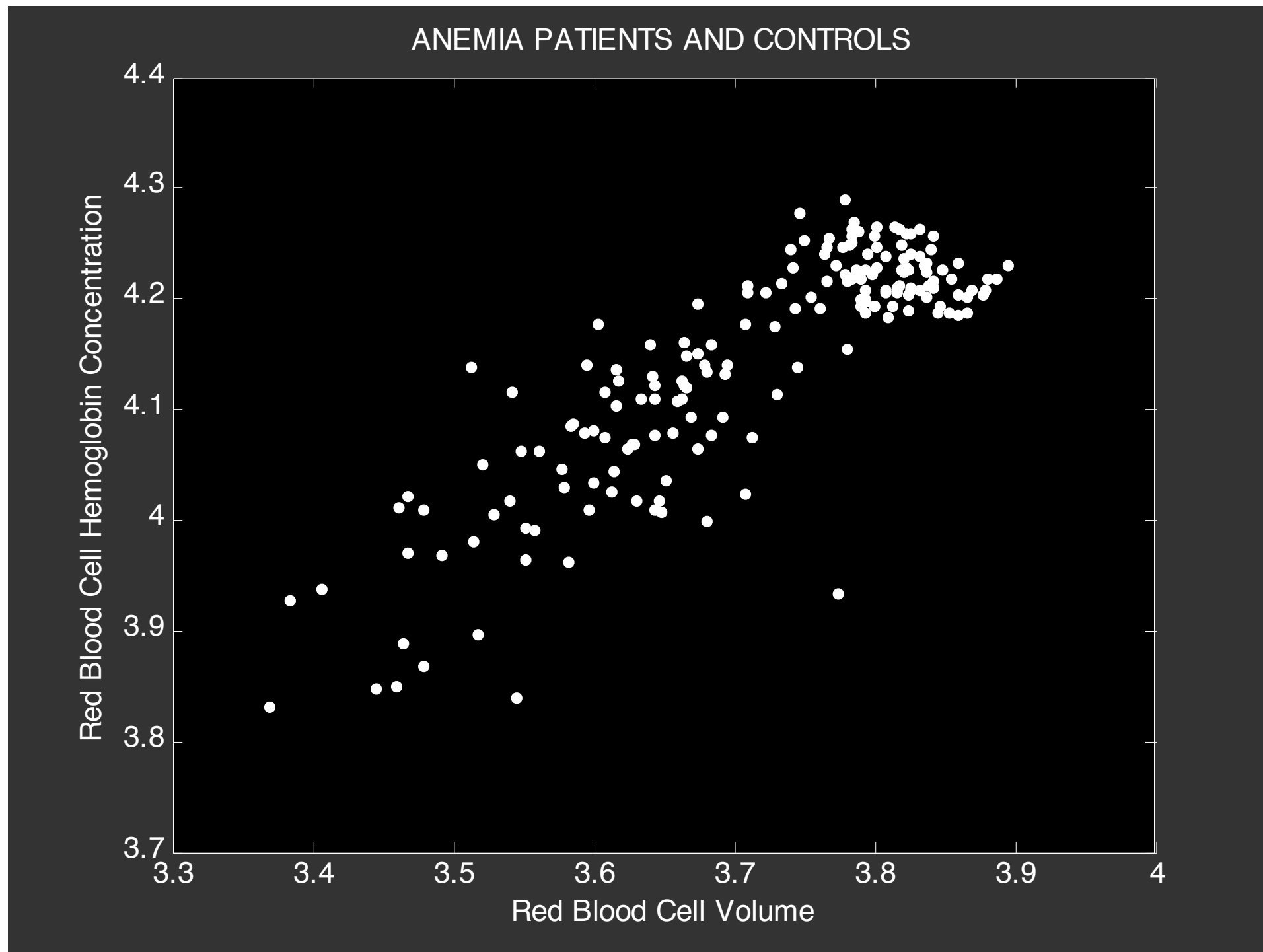


M-step



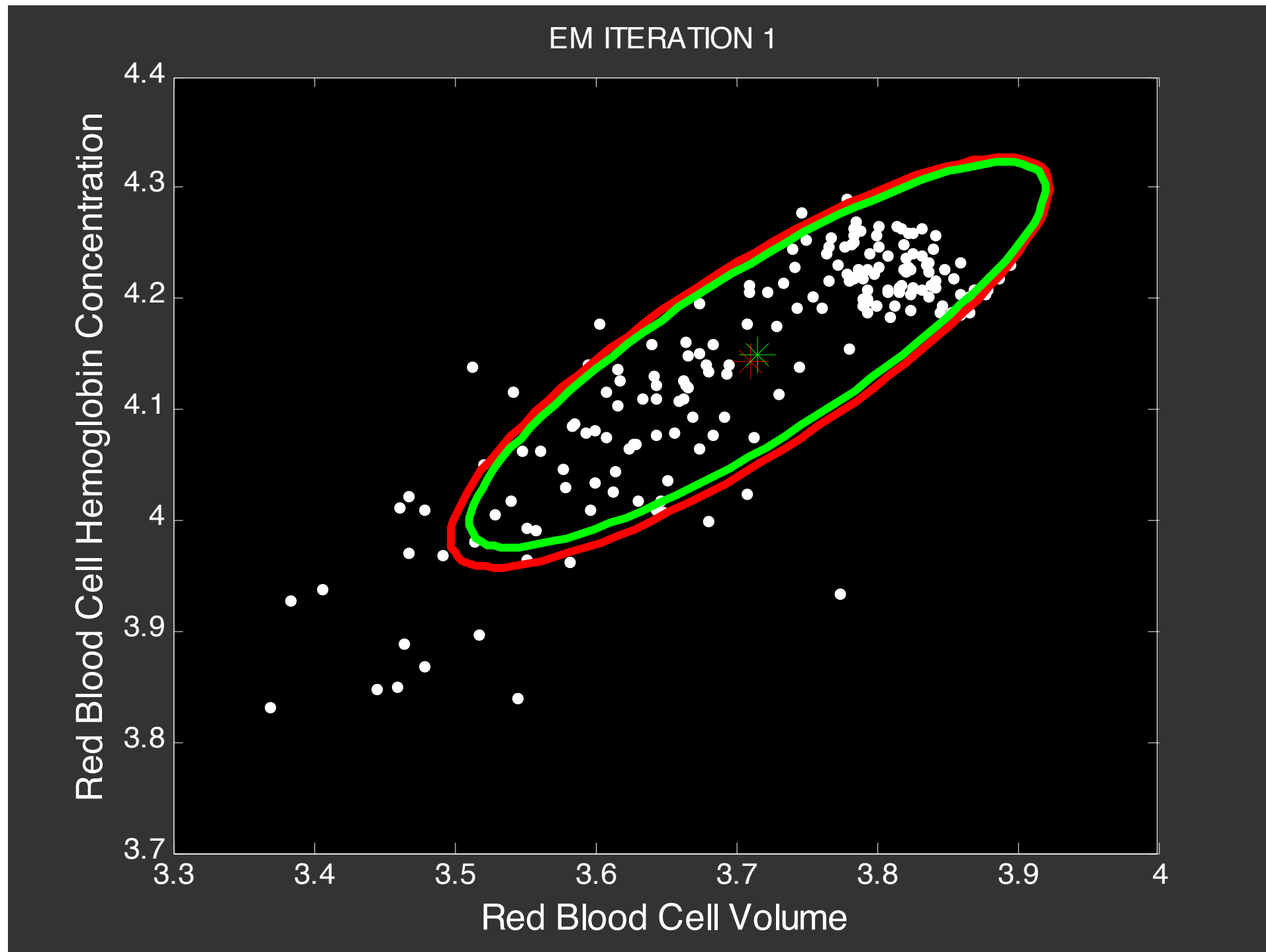


# EM Algorithm for GMM



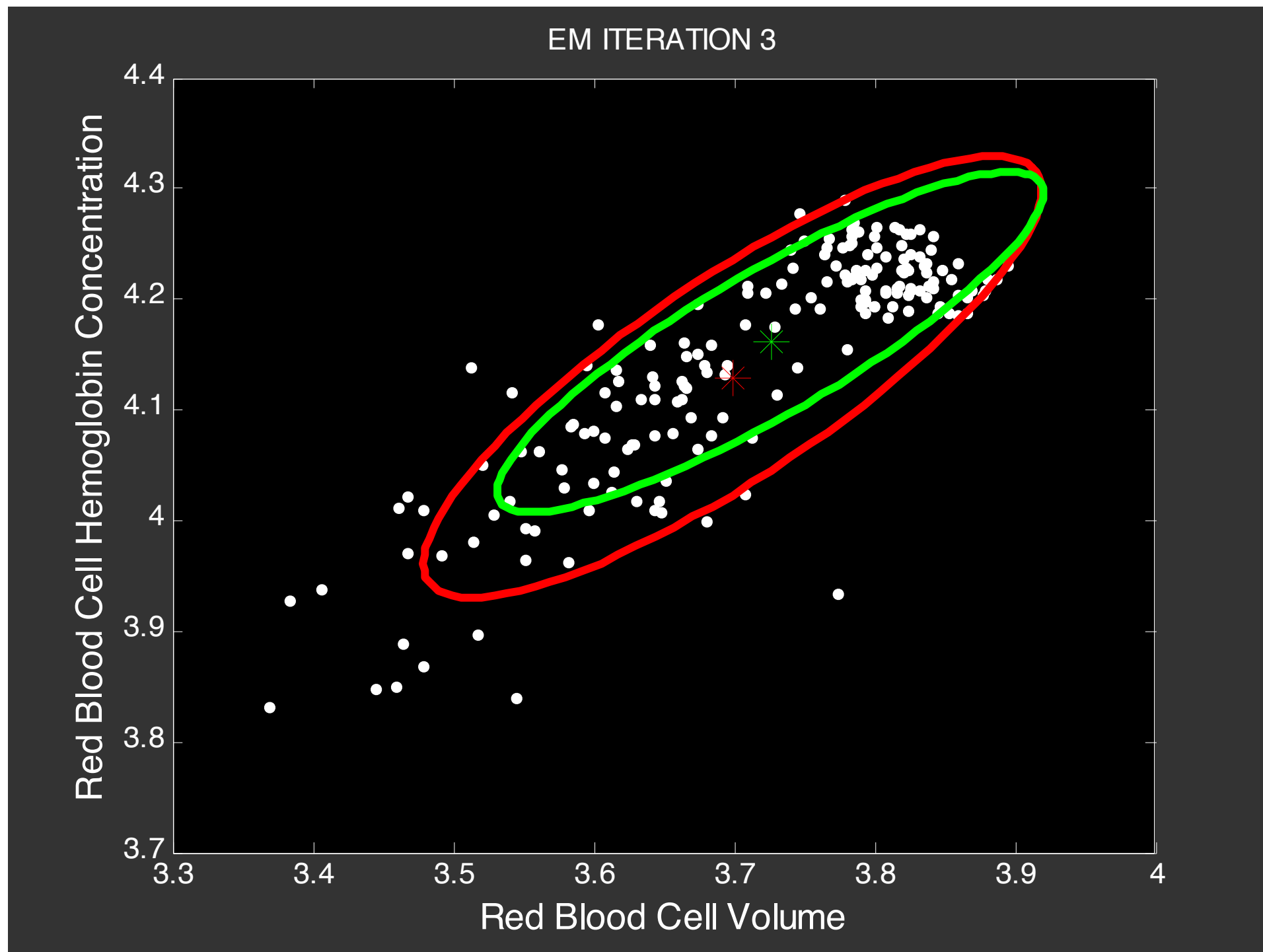
Cadez, Igor V., et al. "Hierarchical models for screening of iron deficiency anemia." *ICML*. 1999.

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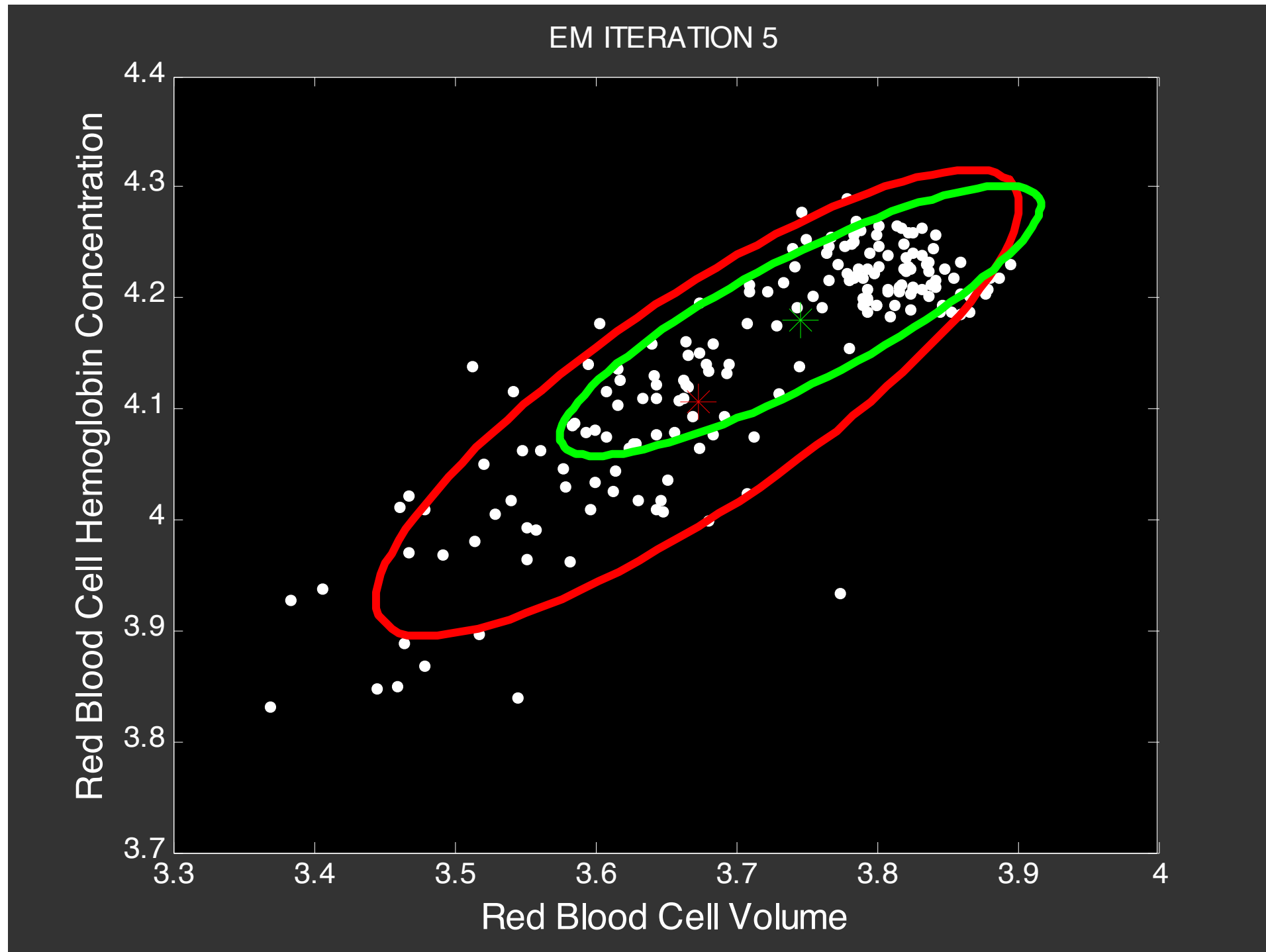
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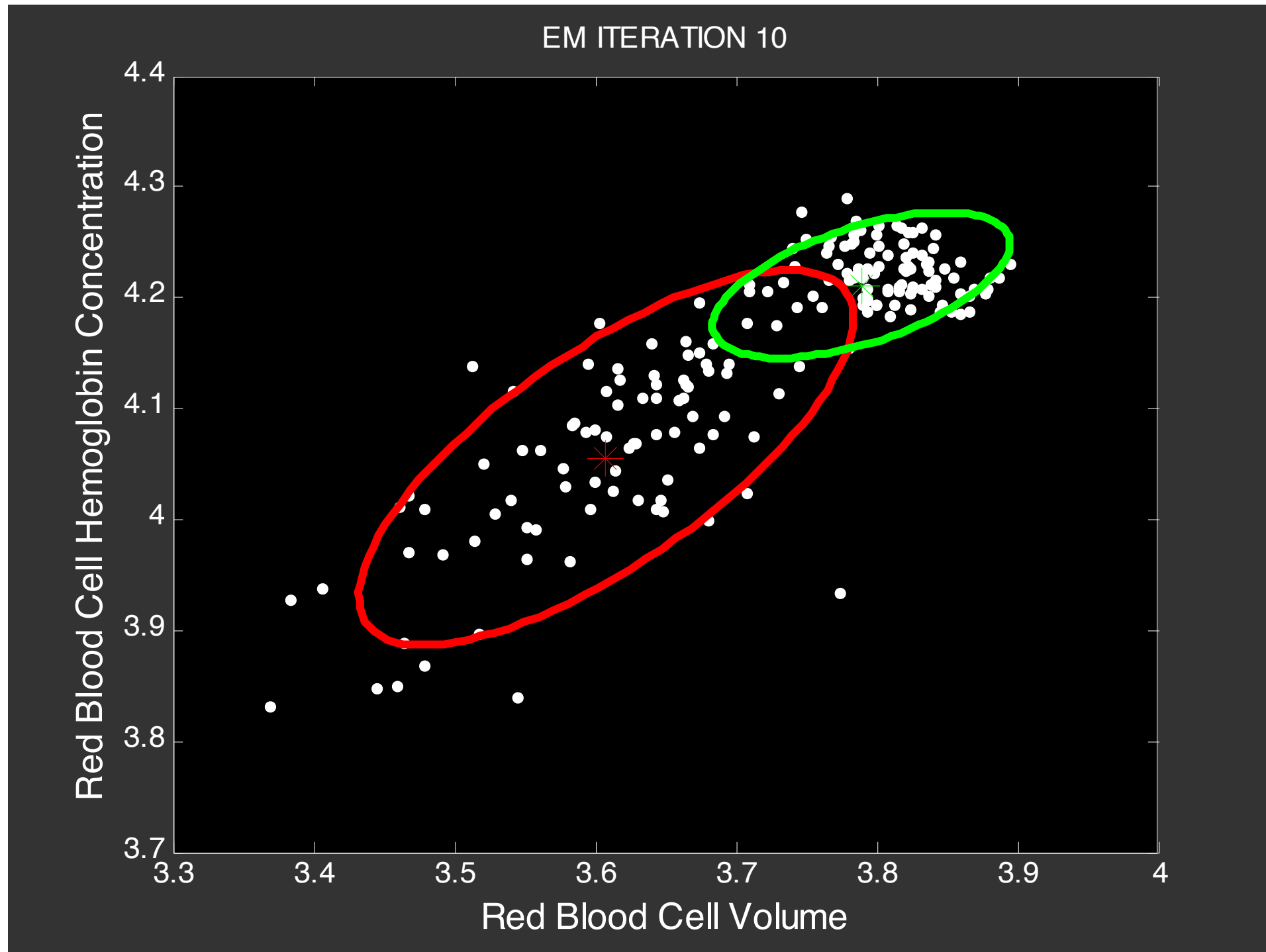
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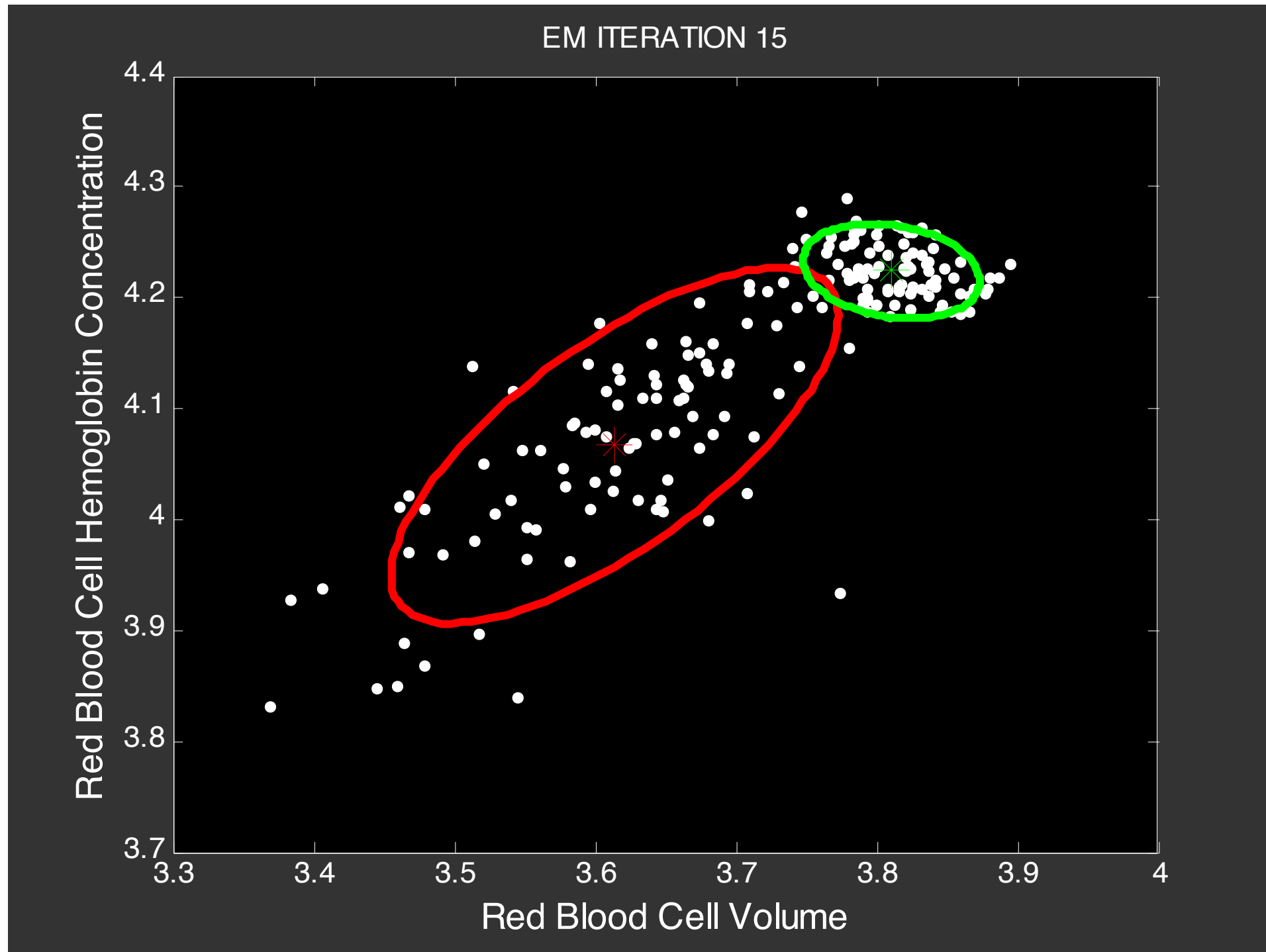
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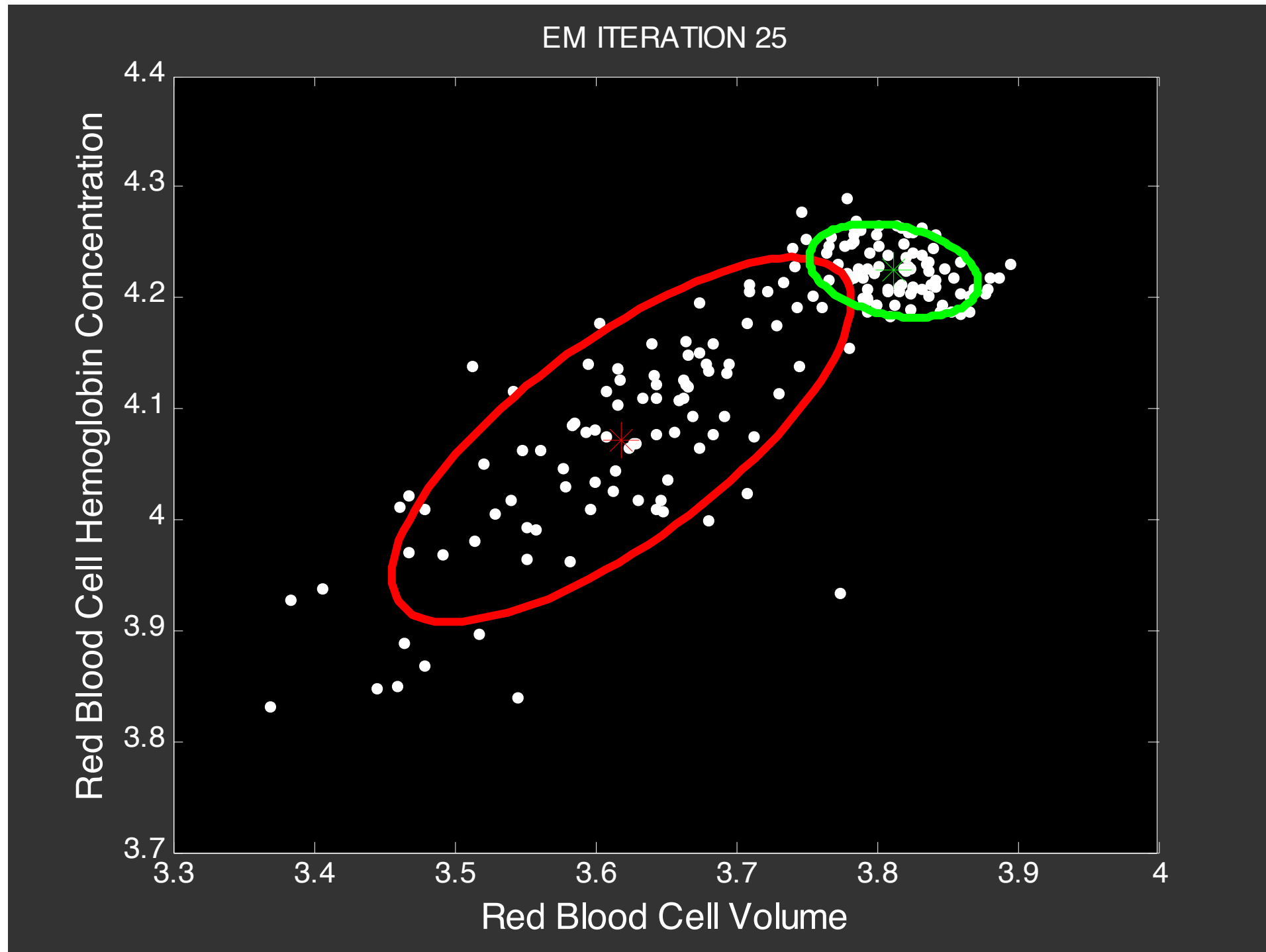


# EM Algorithm for GMM



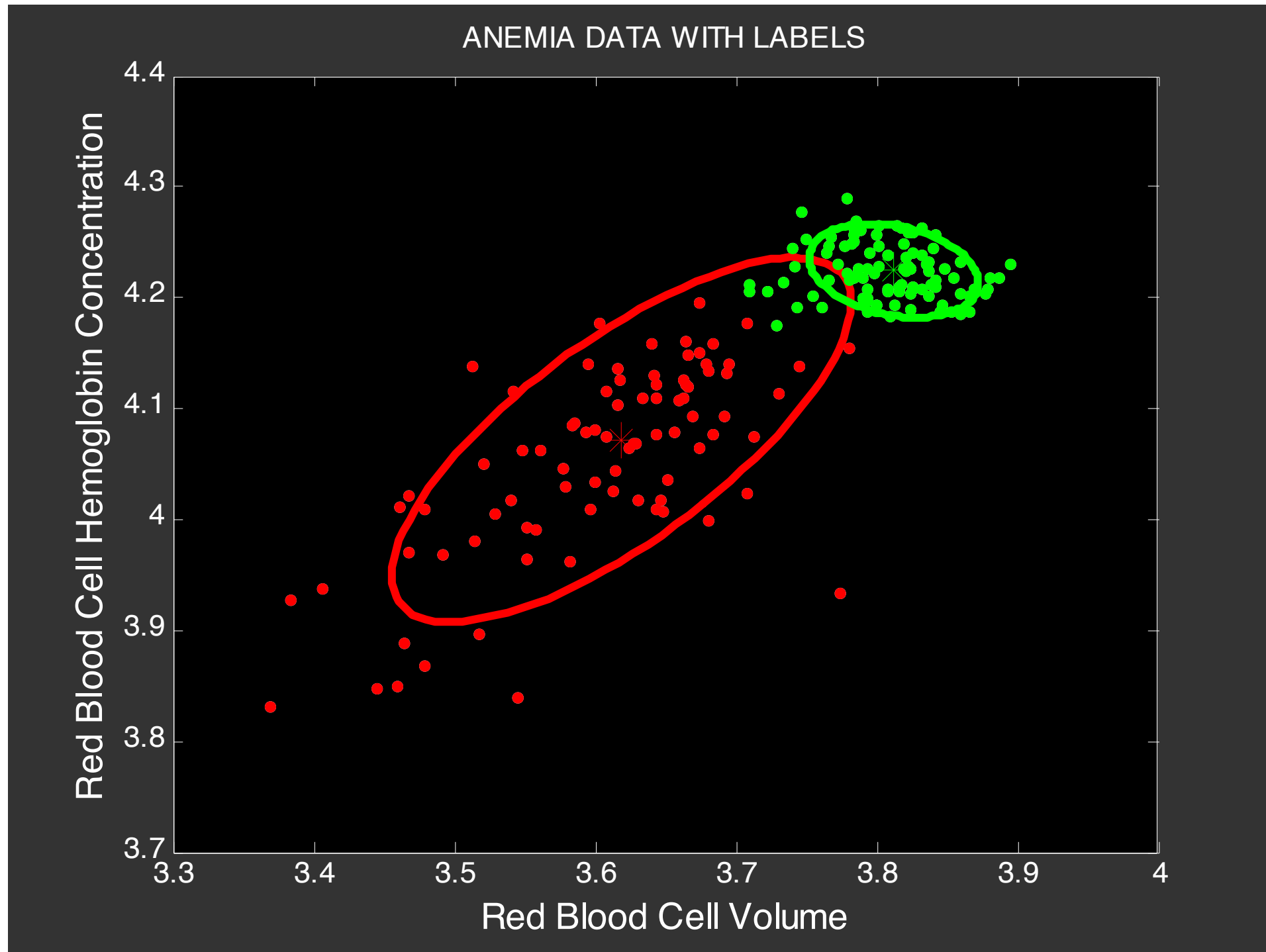
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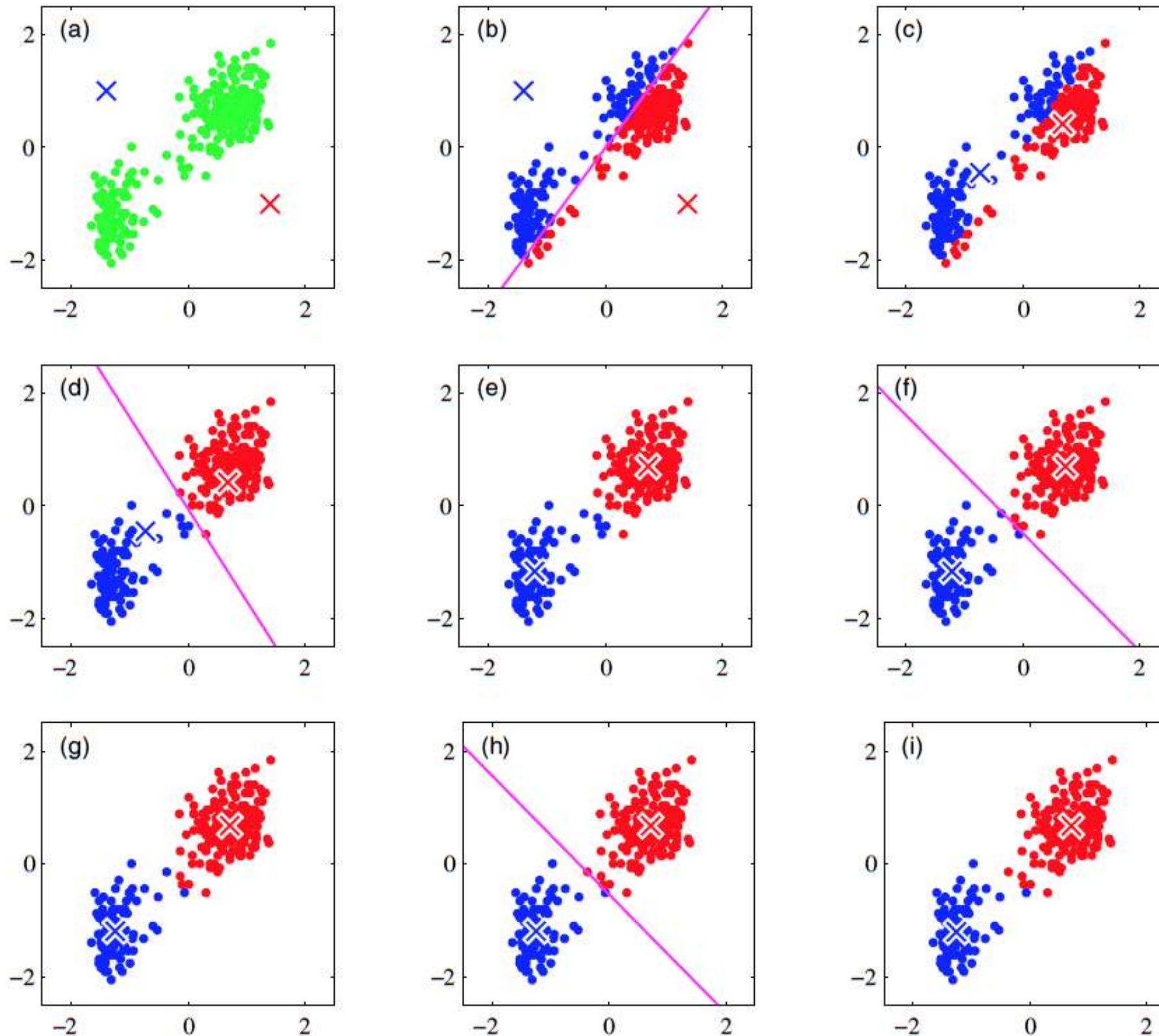
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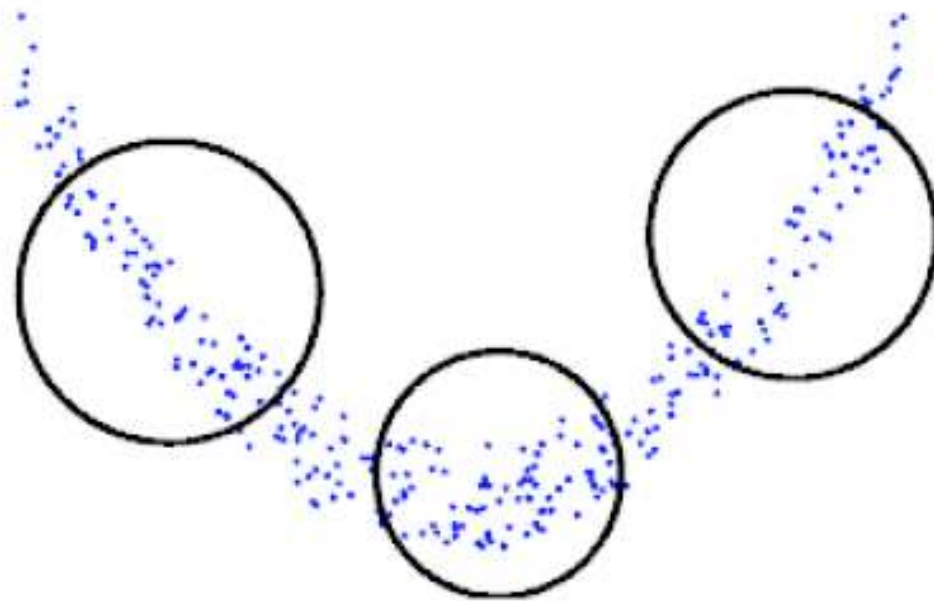
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# K-means Algorithm for Initialization



# Other Considerations

- Initialization - random or k-means
- Number of Gaussians
- Type of Covariance matrix
  - Spherical covariance

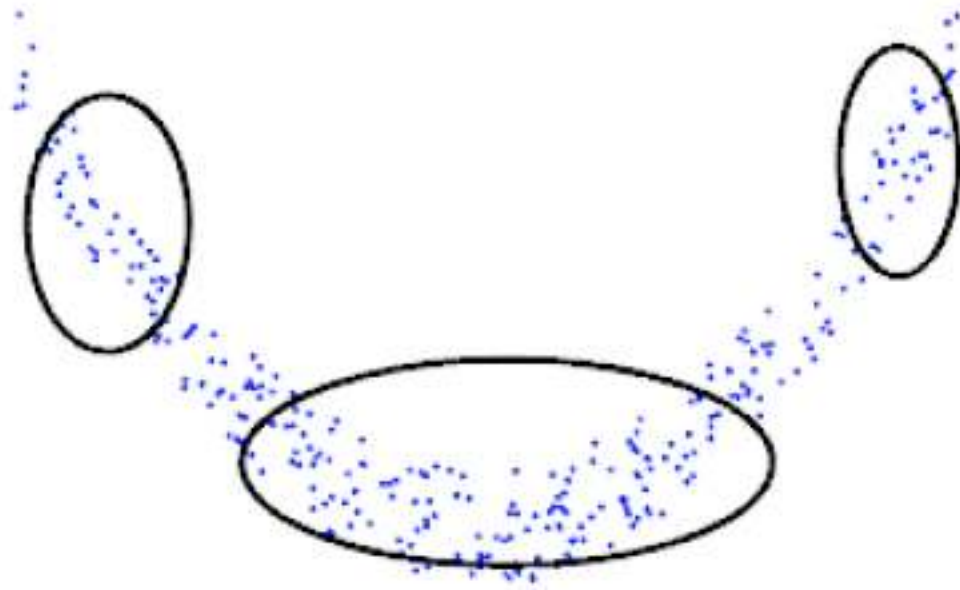


- Less precise.
- Very efficient to compute.



# Other Considerations

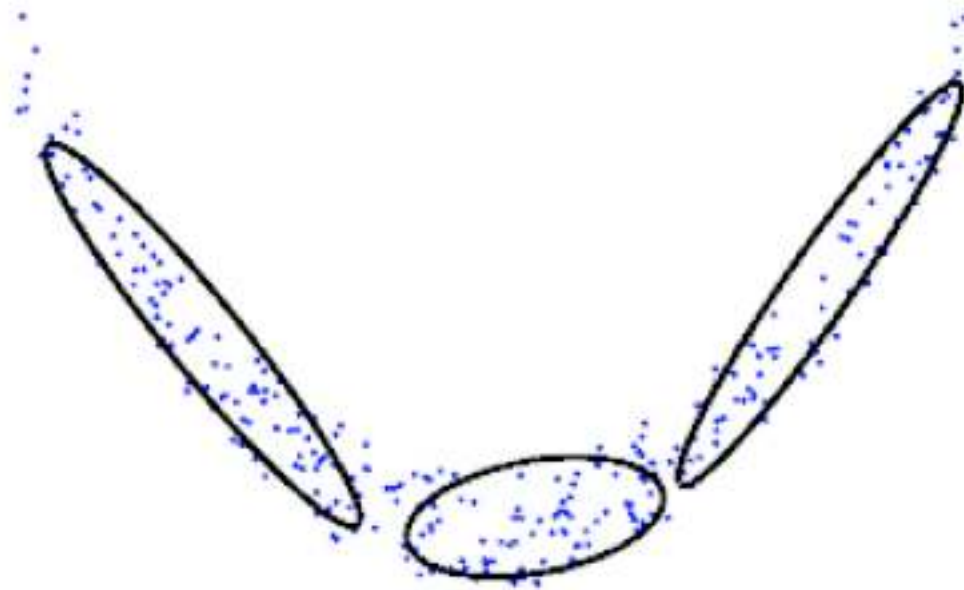
- Initialization - random or k-means
- Number of Gaussians
- Type of Covariance matrix
  - Diagonal covariance



**-More precise.  
-Efficient to compute.**

# Other Considerations

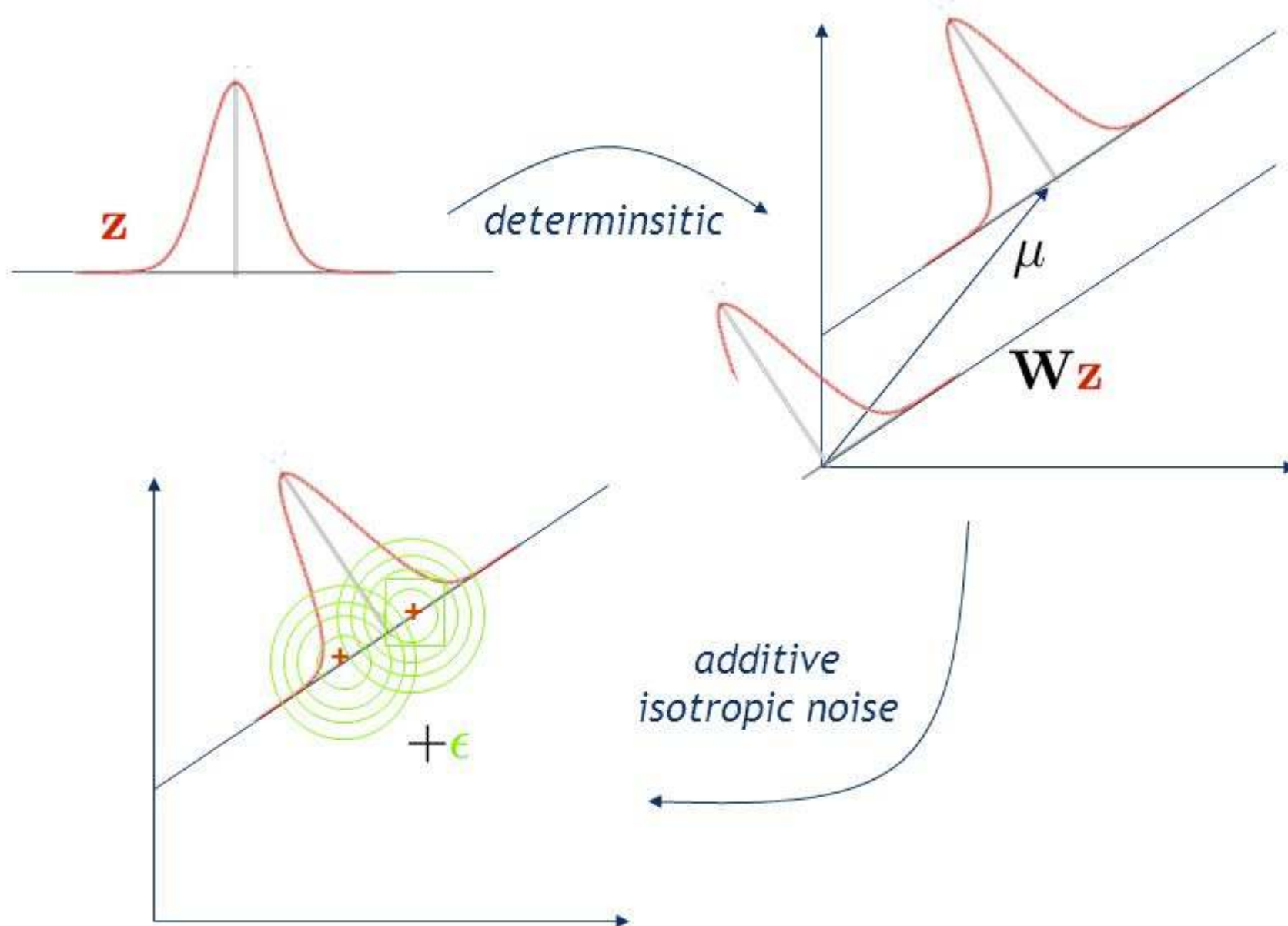
- Initialization - random or k-means
- Number of Gaussians
- Type of Covariance matrix
  - Full covariance



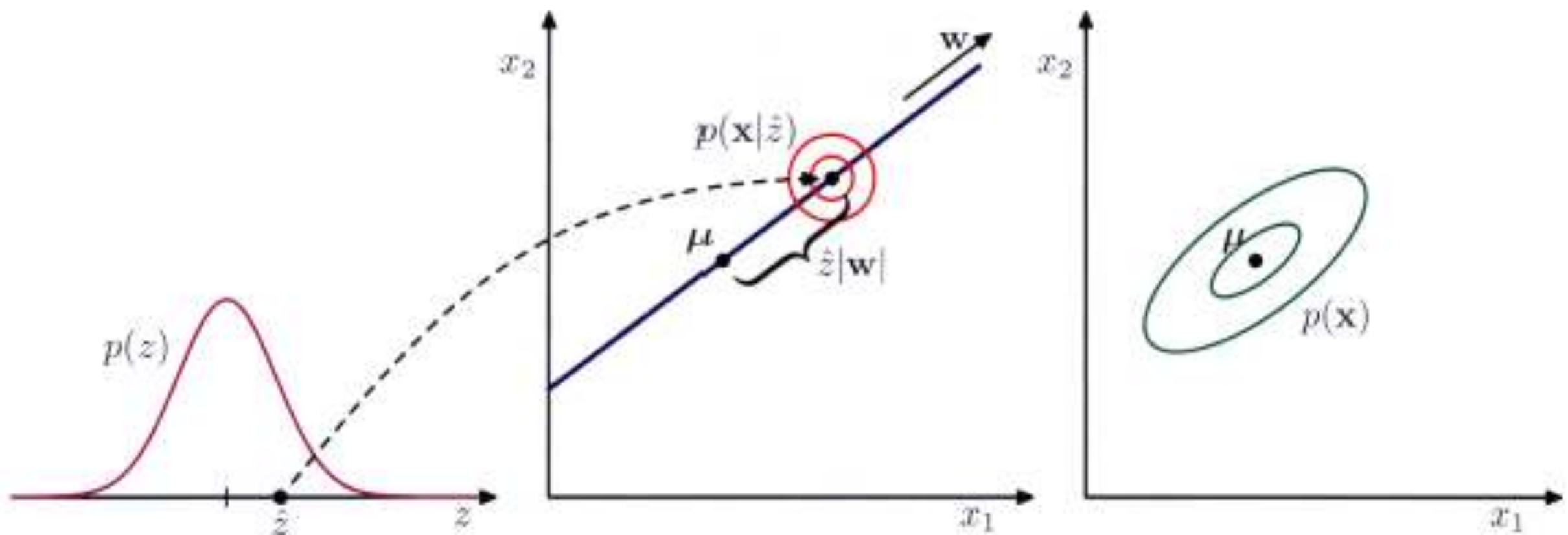
- Very precise.
- Less efficient to compute.

# Probabilistic PCA

## Probabilistic PCA: Illustration



# 1-D PPCA Example



# HouseKeeping

- Assignment 2
- Mid Term Exam
- Project Topic Selection.
- Notes on Assignment 1